# Text-Image-GenAI Master Study Pack

Methods • Evaluation • Ethics

Generated: 2026-05-12 15:51

Sections: 3 | Flashcards: 17 | Key subsections: 3

Use this pack for quick revision, exam prep, and rapid flashcard practice.

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# Methods

Pipeline design, diffusion models, and captioning mechanics

# Text-Image-GenAI — Methods (1■page summary)

Purpose

- Build a modular pipeline that performs text→image generation and image→text captioning, enabling round■trip evaluation and iterative improvement.

Pipeline Overview

- Components: (1) Text encoder, (2) Latent diffusion model (text→image), (3) Vision–language encoder/captioner (image→text), (4) Evaluation & API layer.
- Data flow: Prompt → text embedding → diffusion sampling → generated image → vision encoder → caption → compare to prompt.

Modeling Details

- Latent diffusion: operates in a compressed latent space to iteratively denoise a noise tensor conditioned on text embeddings. Advantages: high fidelity, controllable via classifier■free guidance.
- Vision–language encoders: BLIP/BLIP■2 or similar transformer■based models map images and text into a shared embedding space for captioning and semantic comparison.
- Conditioning & control: use guidance scale and temperature parameters; classifier■free guidance mixes conditional/unconditional denoising to trade fidelity vs diversity.

Training & Objectives

- Multi■objective loss: $L(\Theta) = \alpha \cdot L\_text + \beta \cdot L\_img + \gamma \cdot L\_perc + \lambda \cdot R(\Theta)$.
- L_text: CLIP■style contrastive loss to align text↔image embeddings.
- L_img: diffusion reconstruction/denoising loss (mean squared error in latent/noise prediction).
- L_perc: perceptual loss (VGG features) to preserve visual quality.
- $R(\Theta)$: regularization (weight decay, priors).

Evaluation & Round■Trip

- Round■trip: generate image from prompt, caption image, then compute semantic alignment between original prompt and generated caption.
- Metrics: FID (image quality), CLIPScore (semantic alignment), BLEU/METEOR/ROUGE/CIDEr (caption quality).
- Ablations: vary model sizes, guidance scales, and captioner architectures to measure tradeoffs.

Engineering & Deployment

- Modular design: decouple diffusion model, captioner, and API service (FastAPI). Allows swapping components and offline vs cloud runtimes.

- Efficiency: use latent-space sampling, mixed precision and caching of embeddings; batch captioning during evaluation.
- Scalability: expose endpoints for text→image and captioning; worker queue for heavy generation jobs.

Practical Notes

- Compute: diffusion sampling is CPU/GPU heavy; consider smaller checkpoints or low■rank adapters for constrained environments.
- Ethics & Safety (brief): add watermarking, content filters, and human review for sensitive outputs.

References (key models)

- Latent diffusion / Stable Diffusion family; BLIP / BLIP■2 for captioning; CLIP for embedding alignment.

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Generated as a concise, one■page methods summary for exam prep and quick reference.

# Evaluation

Metrics, ablations, and round-trip consistency

# Text-Image-GenAI — Evaluation (1■page summary)

Purpose

- Describe metrics and procedures used to measure model fidelity, semantic alignment, and caption quality in the text↔image pipeline.

Evaluation Strategy

- Round■trip evaluation: prompt → generate image → caption image → compare caption to original prompt.
- Quantitative metrics:
- FID: image quality (lower is better).
- CLIPScore: semantic alignment between image and text (higher is better).
- BLEU / METEOR / ROUGE / CIDEr: caption/text similarity metrics.
- Perceptual metrics: LPIPS, human eval scores.

Design of Experiments

- Ablation studies: vary guidance scale, model size, captioner architecture, and data augmentation.
- Round■trip consistency: measure semantic drift between prompt and caption (CLIP distance).
- Robustness: evaluate across prompt styles, object composition, and out■of■distribution prompts.

Statistical Reporting

- Report mean ± std over multiple seeds; provide significance testing for ablation differences.
- Use precision/recall for object presence and retrieval tasks where applicable.

Practical Notes

- Use balanced datasets and sampled prompts to avoid metric bias.
- Complement automatic metrics with human rating studies for realism and alignment.

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One■page evaluation summary suitable for exam prep and quick reference.

# Ethics

Bias, misuse, privacy, and mitigation strategies

## Text-Image-GenAI — Ethics (1■page summary)

Core Ethical Considerations

- Misinformation & deepfakes: generated images can be misused to deceive.
- Bias & fairness: training data biases can produce stereotyped or harmful outputs.
- Privacy: models trained on scraped data may inadvertently reproduce personal data.

Mitigations

- Content filters: block harmful prompts and outputs.
- Watermarking: embed detectable watermarks to signal synthetic origin.
- Human review & moderation for sensitive outputs.
- Dataset curation: apply provenance checks and bias audits.

Deployment Considerations

- Rate limiting and monitoring for misuse.
- Provide clear user guidance and disclaimers for generated content.
- Implement reporting and takedown processes.

Research Directions

- Robust watermarking and provenance verification.
- Better bias metrics and debiasing pipelines.

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One■page ethics summary for exam prep and quick reference.

# Flashcards Appendix

Combined question-and-answer review cards

# Flashcards Appendix

## Methods

- Q1: What are the main components of the Text■Image■GenAI pipeline?
- A: Text encoder, latent diffusion model (text→image), vision–language captioner (image→text), evaluation/API.

- Q2: What advantage do latent diffusion models provide?
- A: Operate in compressed latent space for high fidelity sampling and computational efficiency compared to pixel■space diffusion.

- Q3: What is classifier■free guidance?
- A: A technique that mixes conditional and unconditional denoising to trade fidelity vs diversity via a guidance scale.

- Q4: Name the multi■objective loss components used.
- A: L_text (CLIP contrastive), L_img (diffusion reconstruction), L_perc (perceptual), $R(\Theta)$ (regularization).

- Q5: What metric measures semantic alignment of image and text?
- A: CLIPScore.

- Q6: Which metric evaluates image quality?
- A: FID (Fréchet Inception Distance).

- Q7: How is round■trip evaluation performed?
- A: Generate image from prompt → caption the image → compare caption to original prompt (semantic metrics).

- Q8: Two engineering strategies to reduce compute cost?
- A: Use latent■space sampling, mixed precision, caching embeddings, and smaller checkpoints or adapters.

- Q9: Name one safety control to add in deployment.
- A: Content filters, watermarking, and human review.

## Evaluation

- Q1: What is round■trip evaluation?

- A: Generate image from prompt → caption image → compare caption to original prompt (semantic metrics).

- Q2: Which metric measures image quality?
- A: FID (Fréchet Inception Distance).

- Q3: Which metric measures semantic alignment between image and text?
- A: CLIPScore.

- Q4: Name one perceptual metric used for image similarity.
- A: LPIPS.

- Q5: Why complement automatic metrics with human studies?
- A: Automatic metrics don't fully capture realism, coherence, or bias; humans provide qualitative judgment.

## Ethics

- Q1: Name three ethical risks of text→image models.
- A: Misinformation/deepfakes, bias/stereotyping, privacy leaks.

- Q2: One technical mitigation for detecting synthetic images?
- A: Watermarking or provenance metadata.

- Q3: Why is dataset curation important?
- A: Reduces bias, improves provenance, and lowers privacy risks.